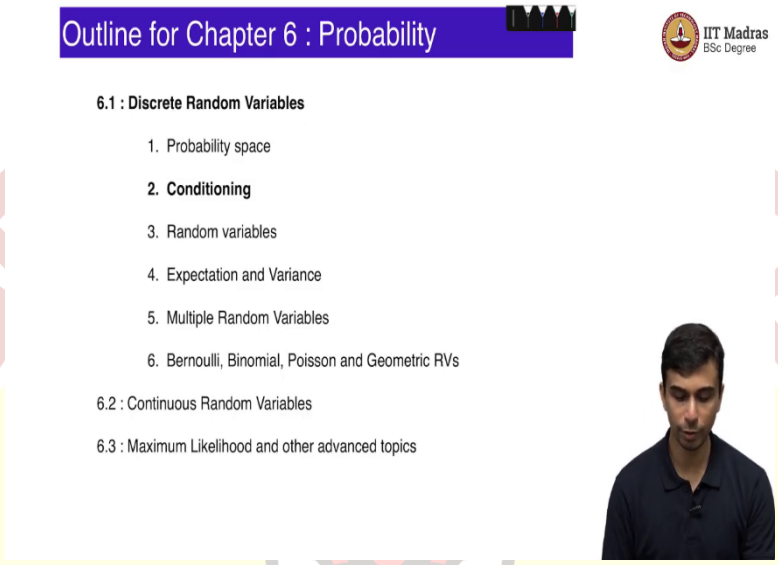


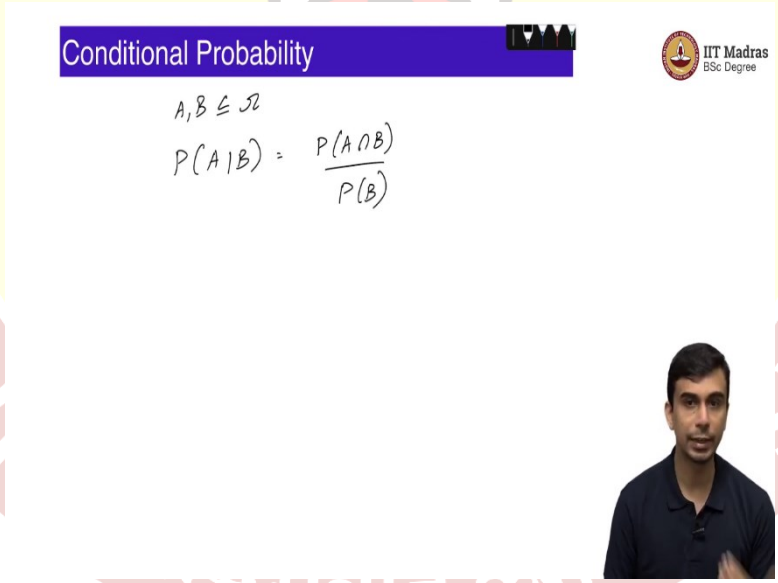
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Assistant Professor Harish Guruprasad Ramaswamy
Department of Computer Sciences & Engineering
Indian Institute of Technology Madras
Conditioning

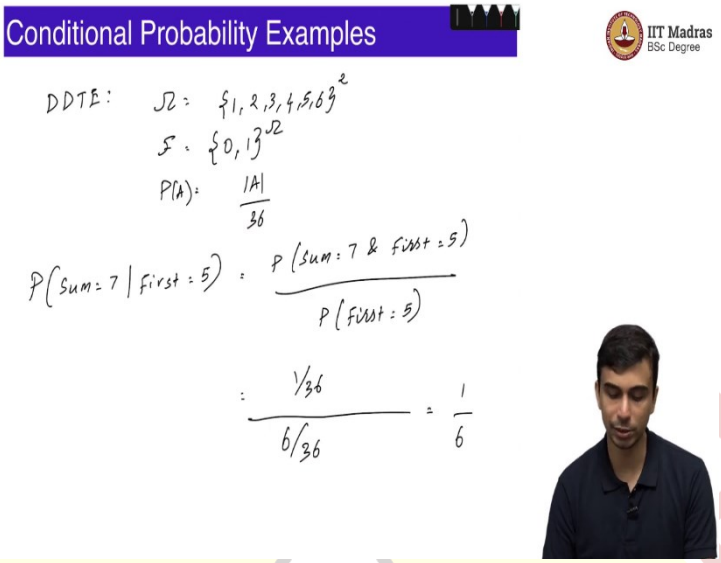
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Hello, everyone. Now that we have the basic tools of probability, namely, the probability space, which consists of the sample space omega under sigma algebra F and the probability measure P, we move on to the next basic element, which is conditioning. We have two elements. Now, we will assume that we are sample space omega and we have two events A , B, let us say A and B are subsets of omega. The conditional probability of A given B is simply defined us probability of A intersection B, divided by probability of P. This is a very intuitive meaning.

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For example, let us say, let us consider the double dice throw experiment. Omega is simply 1, 2, 3, 4, 5, 6 over 2 and F this 0 , 1 power omega. And let us use the probability measure given by the standard probability of A is cardinality of A by 36.

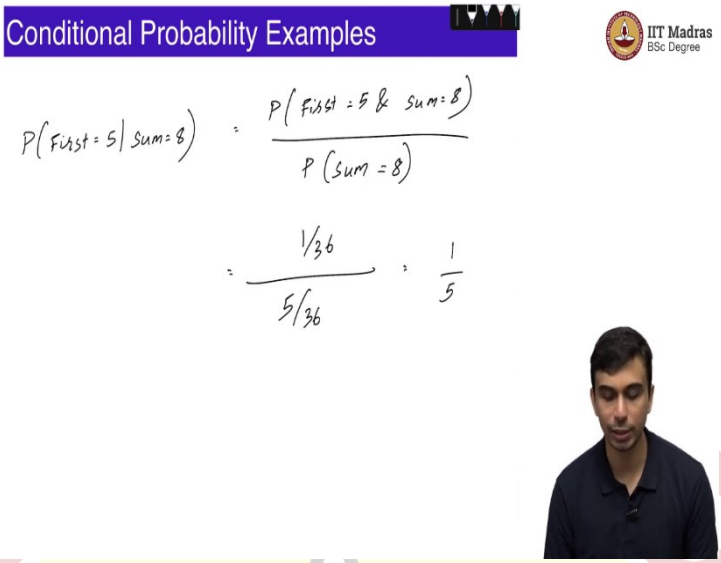
So, in this double dice throw experiment, I can ask the question, what is the probability of the sum of the dice being 7 given that the first die throw equal to 5? What is the probability of the sum of the, well, so this using the definition of the conditional probability this is equal the probability of sum equal to 7 and first equal to 5 divided by probability of first equal to 5.

So, what is the probability of sum equal to 7 and first equal to 5? So, in this case, we are having a counting probability measure, which means, that we just need to count how many of the 36 events satisfy this condition. So, this is clearly sum equal to 7 and first equal to 5 means that 5 , 2 is the only event here. So, the number of outcomes here is 1, so it is 1 by 36.

What about the denominator? What is the probability with that first equal to 5? How many outcomes are there where the first number die throw is 5, well, that is 5 , 1, 5 , 2, 5 , 3, 5 , 4, 5 , 5 and 5 , 6, so there are 6 outcomes, all of which are equally likely. So, we 6 by 36 is the probability of the denominator. So, we have 1 by 6 as the answer here.

So, conditional probability of the sum being 7 given first equal to 5, we know the answer for this. The probability of sum equal to 7 given the first equal to 5 is exactly equal to the probability that the second equal to 2, which is equal to 1 by 6, but still it is nice to see it working out from first principles.

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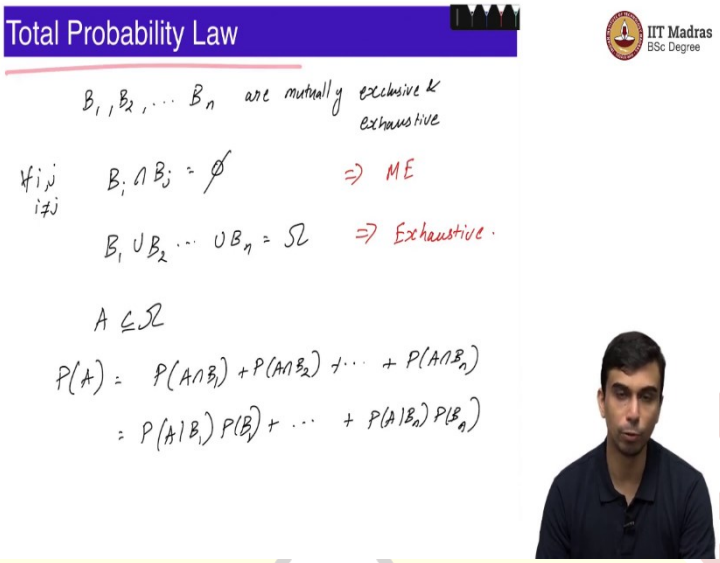


Let us do one more example. Let us do one more example where what is the probability of first equal to 5, given sum equal to 7? This is our let us make, like, but sum equal to 8? We can use the same definition here, which is probability of first equal to 5 given sum equal 8 is equal to the probability of first equal to 5, and sum equal to 8 divided by probability of sum equal to 8.

Because we have a counting probability measure, the numerator is given by the number of favorable outcomes divided by 36, which is first equal to 5 and the sum equal to 8 means that first equal to 5 and second equal to 3, which is only one outcome so it is 1 by 36. And denominator's probability of sum equal to 8. The number of outcomes where the sum is equal to 8 are simply 2 , 6, 3 , 5, 4 , 4, 5 , 3, and 6 , 2, so that is 5, 5 by 36, which is equal to 1 by 5.

So, you can easily see that this is a very basic idea and it in English also, it makes sense. Probability of A given B is the probability of A given that B has already happened. So, you know that B has happened what is the probability that A is happening? What is the probability that A will happen? So, that is the question that conditional probability tries to answer. And these are the kind of basic queries that you should be able to answer using probability.

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So, we can use our axioms of probability along with conditional probability to get what is often called the probability law. Let us say, you have events B1, B2, ., ., . Bn are mutually exclusive and exhaustive. We will define what that means, mutually exclusive and exhaustive. So, you have a collection of events are mutually exclusive, if Bi intersection Bj is equal to null. So, you take any pair of events, the intersection is null, so that, this is for all i , j, i not equal to j, obviously. This is the definition for mutual mutually exclusive.

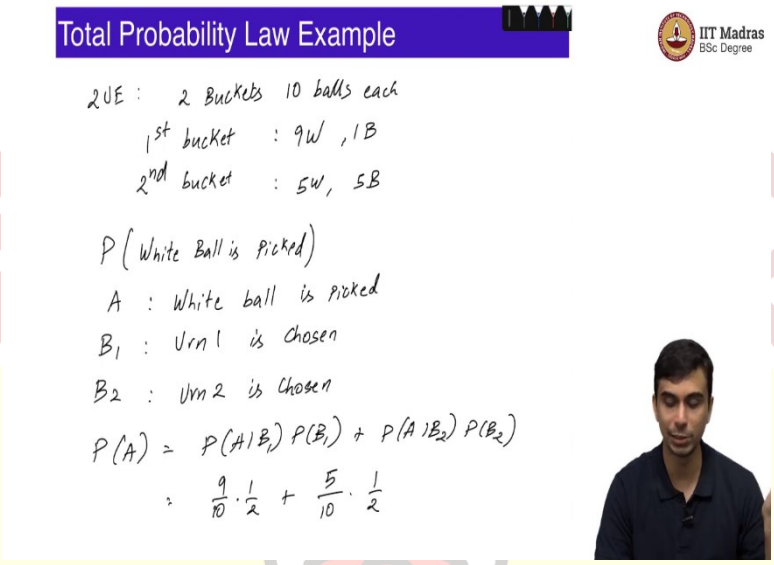
What does exhaustive mean? Well, B1 union B2, union Bn, should be equal to the entire sample space, that is an exhaust, this is the definition for exhaustive. This is definition for mutually exclusive, this is the definition for exhaustive. So, if you have a collection of events, which are mutually exclusive and exhaustive B1 to Bn then you can, let us say you have another event A, A is another event, omega.

We can write probability of A, as equal to probability of A intersection B1 + probability of A intersection B2, + ., ., ., + probability of A intersection Bn. You can do this because each of these are disjoint sets because they are mutually exclusive. B is a mutually exclusive and the union of this should be exactly equal to A because B1 union, B2 union Bn is equal to omega. So, this follows from the axioms of probability and the definition of mutually exclusive and mutually exhaustive.

And you can just undo the not undo, unpack the definition of conditional probability and say probability of A intersection B1 is equal to probability of A given B1 into probability of B1. So, probability of A given B1, would be probably of A intersection B1 by probability of B1, so probability of A intersection B1, you can write as probability of A given B1 into probability

of B1. And so on, you can say probability of A given Bn into probability of Bn. So, this is another often used simple rule, which is the total probability is called the total probability law. It is so widely used that you will not even be aware that you are using this in in practice. For example, let us have a very simple experiment.

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Let us call that the 2-urn experiment. So, you have urn as basically a bucket, let us say, as a bucket. And so, you have two buckets containing 10 balls each, containing 10 balls. So, one, the first bucket contains 9 white balls and 1 black ball, and the second bucket, let us say, contains 5 white balls, and 5 black balls.

So, now, I want to know, let us say, this is the experiment that happened. The experiment that happens is first choose a bucket at random either first bucket or second bucket equally like you toss a coin or let us not bring other randomness you bring you choose one of the buckets equally with equal probability. And then after that, you pick a ball, so that is what happens. And you want to know what is the probability that a white ball is picked.

So, immediately, this is not straightforward to see because there are two buckets and it is complicated, but you can use the total probability law to get this particular thing. So, let us say, A be the event that corresponding to white ball is picked, B1 be the event urn 1 is chosen. Urn 1 or bucket 1 is chosen. B2 is the event that bucket 2 is chosen.

We can clearly see that B1 and B2 are mutually exclusive and mutually exhaustive. So, you cannot choose, your own choosing only one of the buckets so it is either B1 or B2 and you are choosing definitely choosing one of the buckets. So, B1 union B2 should be exactly equal to

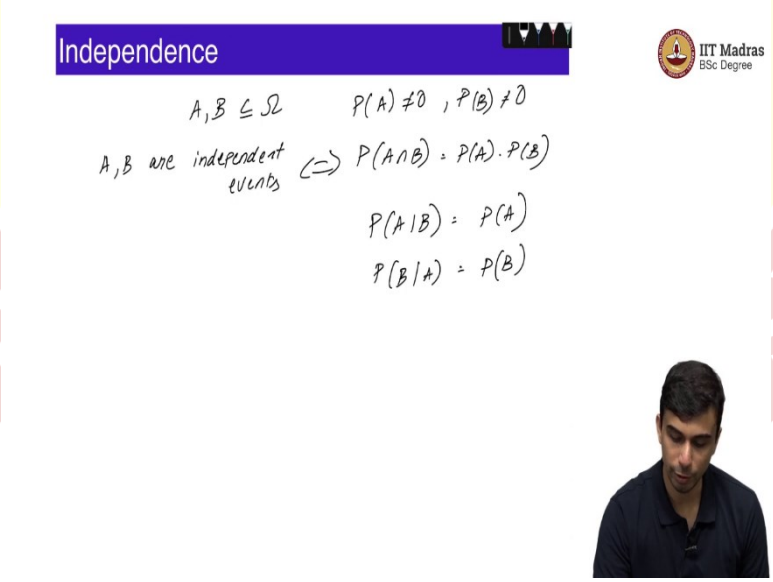
you have to choose one of the buckets, so that is the, so B1 and B2 are mutually exclusive and mutually exhaustive.

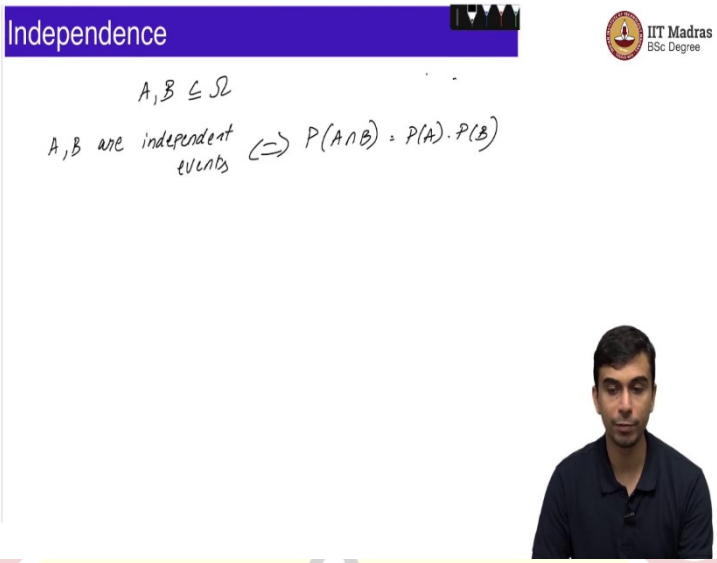
So, you can write probability of A, as probability of A given B1 into probability of B1 + probability of A given B2 into probability of B2. So, what is probability of B1 and probability of B2? Well, we are choosing both the urns with equal probability, so that is both probably a given probability of B2 are 0.5

But what is probably A given B1? Here is where you can use easily you can say, well, given that the first bucket is chosen, what is the probability of the white ball that is 9 by 10? So, you have 9 white balls and 1 black ball. So, that is 9 by 10, into 1 by 2, what is the probability that the ball chosen is white given that the second bucket was picked that is 5 whites and 5 black so it is 5 by 10 into 1 by 2.

So, this is a simple application of the total probability law, and you can see that it is commonsensically, it makes a lot of common sense. And in, the more you are familiar with that, you will actually be using the total property law without even being aware that you are using the total property law.

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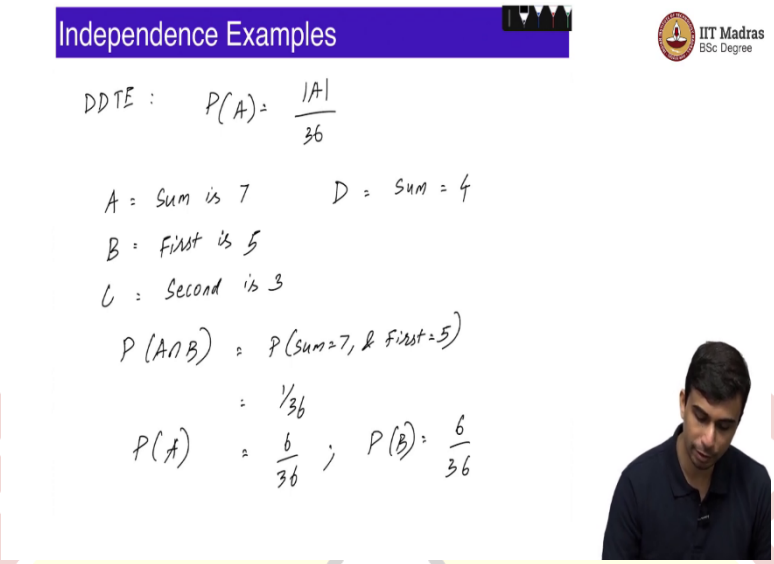


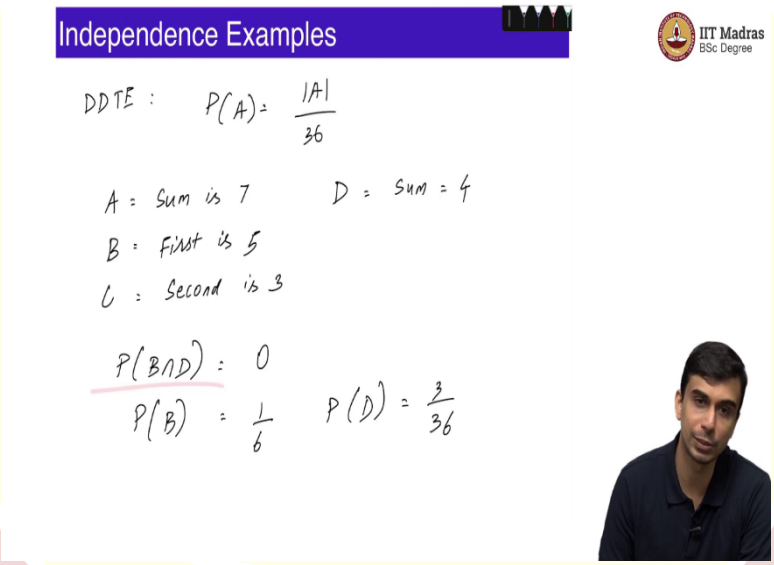
So, now we will move on to the next concept, which is independence. You can have two events A and B, let us say A and B are a subset of the sample space omega A and B are independent, are called independent events. Definition for this is that probability of A intersection B is equal to probability of A into probability of B or in other words, probability of A given B is equal to probability of A.

Why is that? Because again, effectively divide the numerator and left side, both left side and right side we have probability of B which will be a probability of A and similarly probability of B given A will be probability of B. I am assuming for now, I can divide by a probability of B because probability of B is not 0. So, for now, let us assume that A and B are subset of omega but probability of A not equal to 0 and probability of B not equal to 0.

But the definition would be, I mean, you can have, given have probability of A and probability of B as 0 probability events, in which case, this is the true definition of this probability A intersection B, is equal to probability of A into probability of B. That is the true definition. If probability of A and probability B are not 0, then you can get equivalent definitions, that is probability of A intersection probability of A given B is equal to probability of A and probability of B given A is equal to probability of B.

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So, let us have some simple examples. Let us take the two dice throw experiment a dual double dice throw experiment, and the probability of A simply equal to cardinality of A by 36. Well, let us consider the two events given by let us say A is equal to the event that sum is 7 and B is the event let us say first dice throw is let us say 5. C is the event that second dice throw is 3.

I can ask the question are A and B independent? Well, what is probability of A intersection B? What is the probability of A intersection B? It is probability of sum is equal to 7 and the first is equal to 5. Well, that is exactly equal to the probability that first equal to 5 and second equal to 2 which is equal to 1 by 36.

What is probability of A? Well, probability of A is probability of sum equal to 7, that can be shown to be 6 by 36. And probability of B is equal to probability that first is 5, which is once

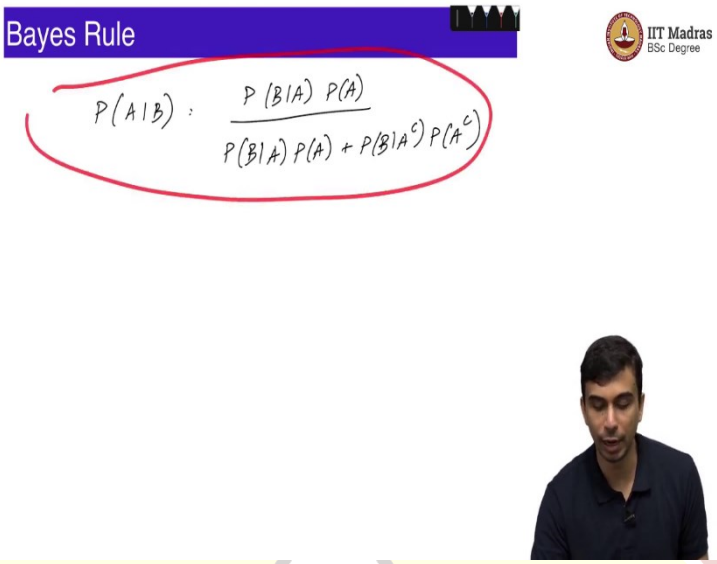
again 6 by 36. So, in this case, it does seem that probability of A intersection B is exactly equal to probability of A into probability of B. Let us now change the, we will have another event and say D is sum equal to 4.

Now, let us use this extra element also. Here we have A and B are independent. And let us take them B and D, are B and D independent events? You can immediately see that B and D are not independent events, but we will come from first principle's again. What is probability of B intersection D? There is this probability that the first is 5, but the sum is 4, can you even think of such an event happening? What is the probability that the first is 5 and the sum is 4? Well, that is, there are no such outcomes that you can think of, so it is actually a null set. So, probability of B intersection D is actually equal to 0.

What is probability of B? Well, probability of B is probability in the first dice throw is 5 that is equal to 5 by 6 by 36, which is 1 by 6. What is the probability of D? The sum is equal to 4. How many outcomes can you think of for that? Well, 13, 22, 31 and so that is 3 by 36. So, we can clearly see that probability of B intersection D is not equal to probability of B into probability of P.

So, we have the A and D are not independent events, but A and B, for example, are independent events. And, similarly, B and C are also independent events. You can, when someone gives you two events, how do you compute whether, how do you say whether they are independent or not? Well, you can just compute probability of A intersection, intersection of those two events and see whether it matches the product of those events. That is the simple definition of independence.

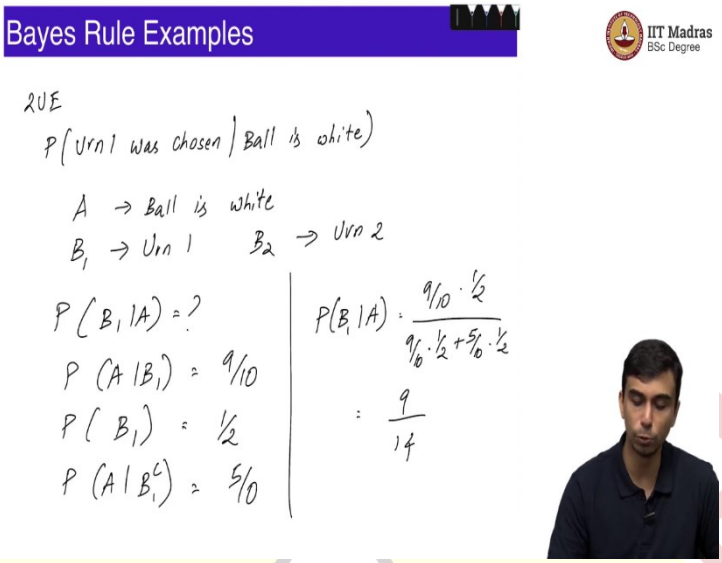
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So, now we move on to another advanced, slightly advanced technique, which is called the Bayes rule. Bayes rule simply gives you a way to flip the conditioning. For example, this is what it says, that is, probability of A given B is equal to probability of B given A into probability of A divided by probability of B given A into probability of A + probability of B given A complement into probability A complement. Why is this true?

Well, the numerator is clearly, you can clearly see this is equal to probability of A intersection B, and the denominator is by the total probability law, exactly equal to probability of B. So, where A and A complement act as the mutually exclusive and mutually exhaustive sets. So, this is the Bayes rule. This is not really new by any words, you are just put together the condition and the definition of conditioning, along with the total probability law to get the Bayes rule. But the Bayes rule is a very powerful and often used tool in machine learning and in general probability for understanding logic itself.

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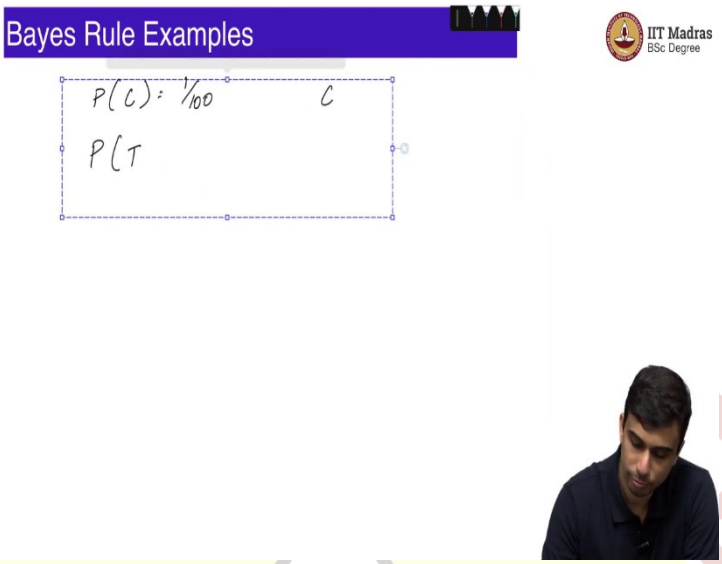
So, let us have some simple examples here. Let us take the 2-urn experiment where you had, remember you had two buckets with the first bucket had 9 white balls and 1 black ball, the second urn had 5 white balls and 5 black balls. You can ask the question probability that the urn-1 was chosen, given the chosen ball was white. This is, you cannot immediately read out the answer to this question, you have to do some calculation for this. And the key tool for doing that is the Bayes rule.

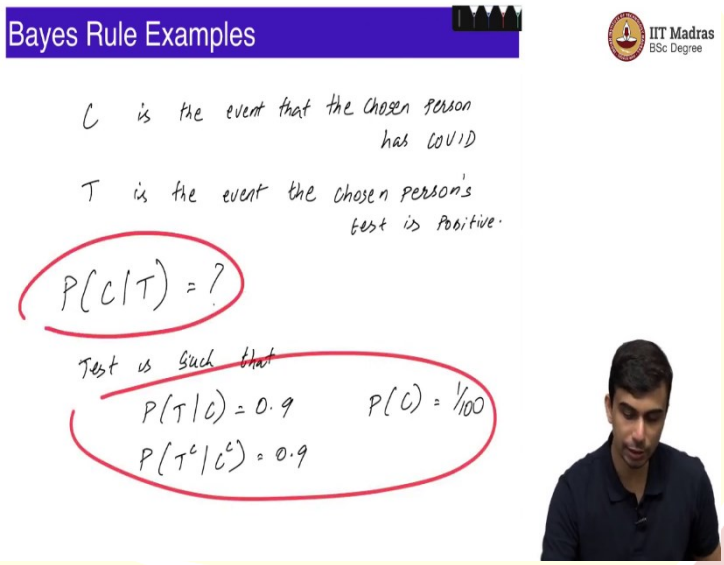
Let us use the same notation that we had earlier that is A corresponds to ball is white and B1 corresponds to first urn us chosen urn-1 is chosen and B2 corresponds to urn-2 is chosen. So, what do we want? We want probability of A given, sorry, probability of B1 given A, this is what we want? Well, that is straightforwardly not easy to see. But what we do here, probability of A given B1 is easy, which is simply equal to 9 by 10.

Because the first urn had 9 white balls and 10 black balls. And what is probability of B1, well that is equal to half. And what is probability of A given B1 complement. So, B1 complement is simply equal to B2 probability of A given B1 complement is simply B2 that is equal to 5 by 10. And what is property of B1 complement, that is also equal to 1 by 2.

You can put all of that together, and get probability of A, sorry, B1 given A simply equal to 9 by 10 into 1 by 2 divided by 9 by 10, into 1 by 2 + 5 by 10, into 1 by 2 which is equal to 9 by 14. So, this is a simple kind of calculus, that simple calculations that you will do for in the Bayes rule, to get to go from A given B to B given A.

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Let us have one more example, which is pretty relevant in this timeline. Let us say, you have 100 people in a room. Let us say you have 100 people in the room and you in that room, you know that the probability of any given person having COVID is let us say, 1 percent. So, let us say probability of COVID is equal to 1 percent, so that is 1 by 100.

And you have, let us say, you have access to an RTPCR test, but this is the situation that is the test being true, let us say T is the event that C is the event that let us make the experiment clear first. C is the event that so this is the experiment, the experiment is, you have 100 people in a room, and you choose a person there and you perform the COVID test the PCR test, and that is the experiment.

So, what are the sample? What is the sample space here? Well, potentially you can have any number of people there can have COVID or not. And based on that, the test can be either positive or negative. So, it is quite complex sample space and sigma algebra, but we will not really be bothering about that, because we do not need to explicitly write down the sample space and the sigma algebra.

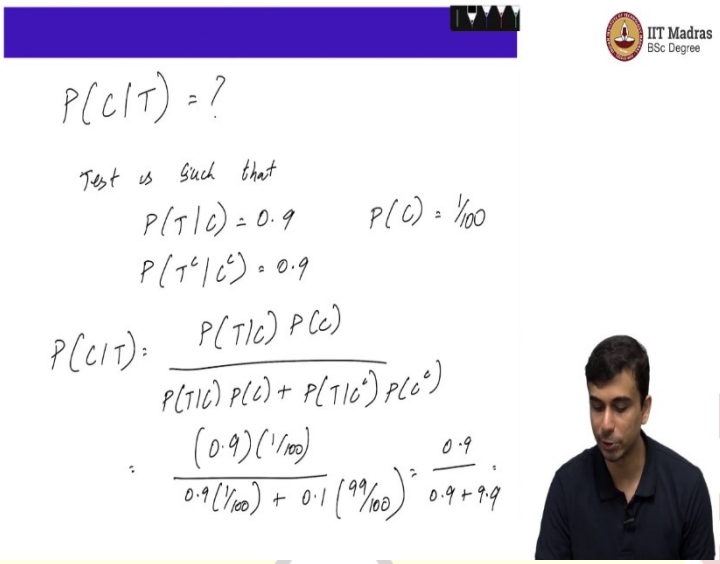
So, we will just give the probabilities that we will be, events that we will be interested. C is the event that the chosen person has COVID. T is the event that the chosen person's test is positive. This is the, now, let us say, you have done the experiment that is you have chosen one person there at random and you have done your PCR test, and it has come out to be positive. And you want to go and speak to the person and say that you have COVID or not.

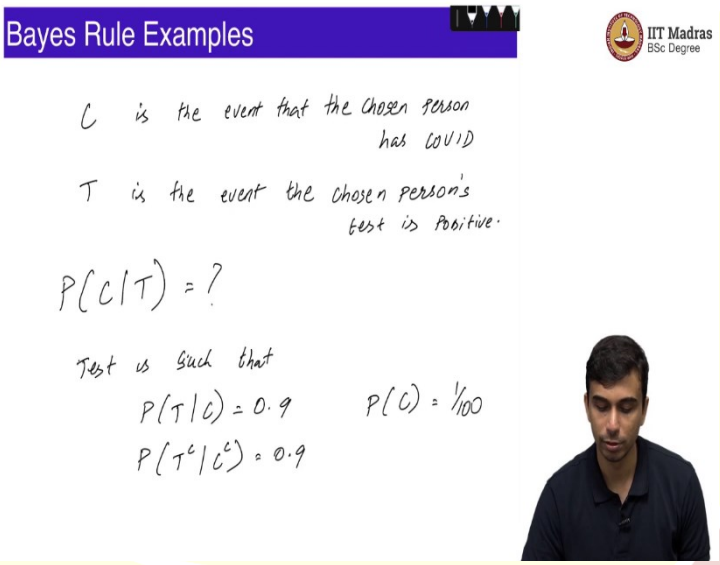
So, what you want to be able to say is, what is the probability of COVID given the test is true? This is what you want. But the test is not perfect, I mean, if you have had a perfect test, then the answer is yeah, the person has COVID. But the test is not perfect. You would see the test is only 90 percent accurate. So, what do I mean by that? Well, the test is such that probability of the test being positive given COVID is 0.9, and the probability of test being not positive, given COVID is not true is also 0.9.

So that is, if you have COVID, it will say 90 percent accurate answer whether you agree with the truth, similarly, even if you do not have COVID, the test will say, whether with 90 percent chance, it will say false. So, that is the, that is what the, this is the characteristic of the test, which is that test manufacturers have done statistics, and this is what they give you that is property of T given C is equal to 0.9 and probably have T complement given C complement is also 0.9.

So, what does the right side mean? That means, that if the tested person truly has COVID, then with 0.9 probability, the test will be positive. If the tested person truly has does not have COVID, then that test will say the test will say negative with 90 percent chance that is what this particular thing saying. And in addition, you also have the extra piece of information that in this particular population of 100 people, one percentage of people have COVID. So, probability of C itself is equal to 1 by 100. So, with these three facts, you want to compute the probability of COVID given.

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Let me just copy this and paste this. Well, you can use the Bayes rule once again here. What is probability of C given T? Well, probability of C given T is equal to probability of T given C into probability of C divided by probability of T given C into probability of C + probability of T given C complement into probability of C complement.

So, every one of these numbers is known here the numerator denominator probability of T given C is 0.9 into 1 by 100 divided by 0.9 into 1 by 100 + probability of what is probably of T given C complement? So, given that problem of T complement given C complement is 0.9 which means that probability of T given C complement is simply 1 minus 0.9 which is equal to 0.1

And what is probability of C complement? Well, probability of C complement is simply 99 by 100. We can do this computation here which is let us say multiply the numerator and denominator by 100 which is 0.9 by 0.9 + 9.9, so it is approximately 0.9 by 100. So, sorry 0.9 by 11. So, you can compute this number which is 0.9 by 0.9 + 9.9 which is approximately about 10 percent.

So, which means, that even though you have discovered, even though your test is positive, the probability that you really have COVID is only 10 percent. So, yeah, it is higher than you when you began with; when you began with you had 1 percent chance of having COVID, but because the test has become positive your chance has increased to 10 percent. But it is key to note that probability of C given T is not probability of T given C. You have 90 percent accurate test is a pretty accurate test, but probably have C given T even with that given that your test has come positive does not mean that you have COVID with 90 percent chance.

Because of the general prevalence being only 1 percent, probability of C given T is not equal probability of T given C, in fact, it is equal to 0.9 by 0.9 + 9.9, which is equal to 0.9 by 10.8 which is around 10 percent which is less than 10 percent. So, this is a simple application of Bayes rule. And with that, we will wrap up the conditioning part of the probability chapter.